Inter-Rater Reliability Between Human Raters & Qwen

Emily Han

January 27, 2025

0.1 Reliability Between Human Raters Within Group

0.2 Human Rating Within Group Reliability

explain why we specifically chose to use Krippendorff's alpha

What does the result mean?

What the result look like the way it is

Table 1: Krippendorff's Alpha Values by Metric and Rater Group

Metric	Group A	Group B	Group C
Tone Appropriateness	0.125	-0.122	-0.174
Emotional Calibration	0.443	-0.220	-0.175
Emotional Escalation	0.345	0.131	-0.108
Functional Empathy	0.035	-0.184	0.033
Contextual Appropriateness	0.380	-0.259	0.147
Emotional Coherence	0.256	0.006	0.193
Emotional Adaptivity	-0.097	0.095	-0.358
Problem Resolution	0.792	0.165	-0.181

0.3 Reliability Between Human Raters Vs LLM

the things that i calculate Explain the spearman rho and what it is interpreting in general Explain the how it relates to the alingment normally and how it is in my specific case

Table 2: Score Distribution Comparison

Score	LLM Ratings (N)	LLM Ratings (%)	Human Ratings (N)	Human Ratings (%)
1	NA	NA	18	2.38095238095238

Table 2: Score Distribution Comparison

Score	LLM Ratings (N)	LLM Ratings (%)	Human Ratings (N)	Human Ratings (%)
2	3	1.19047619047619	81	10.7142857142857
3	2	0.793650793650794	153	20.2380952380952
4	45	17.8571428571429	314	41.5343915343915
5	202	80.1587301587302	190	25.1322751322751
Total Observations	252		756	

0.3.1 LLM & Human Alignment by Metric

Table 3: LLM-Human Alignment Statistics by Metric

metric_id	n	alignment_rate	MAE	Spearman_rho	human_score_avg	llm_score_avg
Tone Appropriateness	45	0.756	1.044	-0.122	3.756	4.756
Emotional Calibration	45	0.689	1.222	-0.283	3.689	4.867
Emotional Escalation	45	0.756	1.067	0.496	3.711	4.644
Functional Empathy	45	0.733	1.133	-0.037	3.844	4.933
Contextual Appropriateness	45	0.733	1.156	-0.059	3.844	4.867
Emotional Coherence	9	0.889	0.778	0.273	3.889	4.667
Emotional Adaptivity	9	0.889	0.778	-0.342	3.667	4.222
Problem Resolution	9	0.778	0.889	0.195	3.889	4.333

0.3.2 LLM & Human Alignment by Scenario

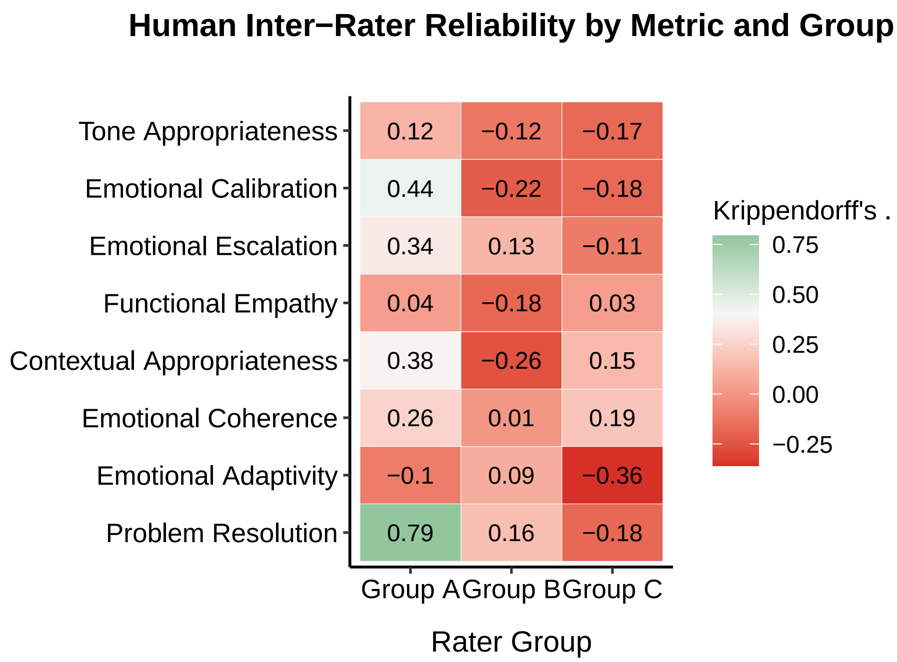

Figure 1: Human Inter-Rater Reliability by Metric and Group

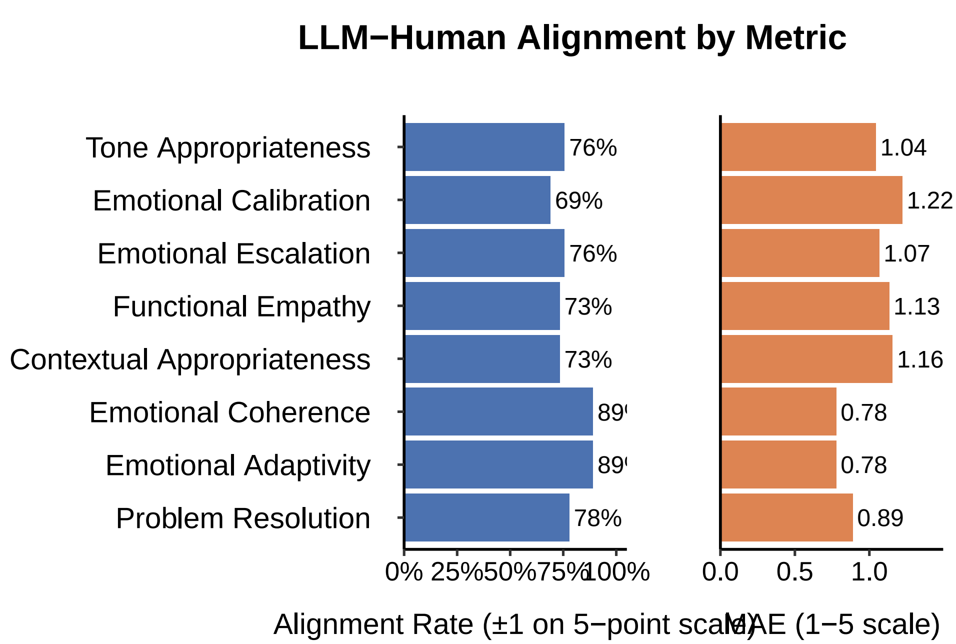

Figure 2: LLM–Human Alignment by Metric

Table 4: LLM-Human Alignment Statistics by Scenario

level	emotion	alignment_rate	Avg_Var	Spearman_rho	MAE
L1	angry	0.8928571	1.4166667	0.1892775	0.8214286
L1	frustrated	0.5714286	0.9285714	0.0193541	1.4285714
L1	neutral	0.9285714	0.8452381	0.1248636	0.6785714
L5	angry	0.5000000	1.5000000	0.3447251	1.6071429
L5	frustrated	0.9642857	0.2380952	0.0377012	0.6428571
L5	neutral	0.9642857	0.4523810	-0.0226356	0.7142857
L10	angry	0.2142857	1.2261905	0.3555958	2.0714286
L10	frustrated	1.0000000	0.7738095	-0.2979218	0.6071429
L10	neutral	0.6785714	0.8928571	-0.1908854	1.2500000

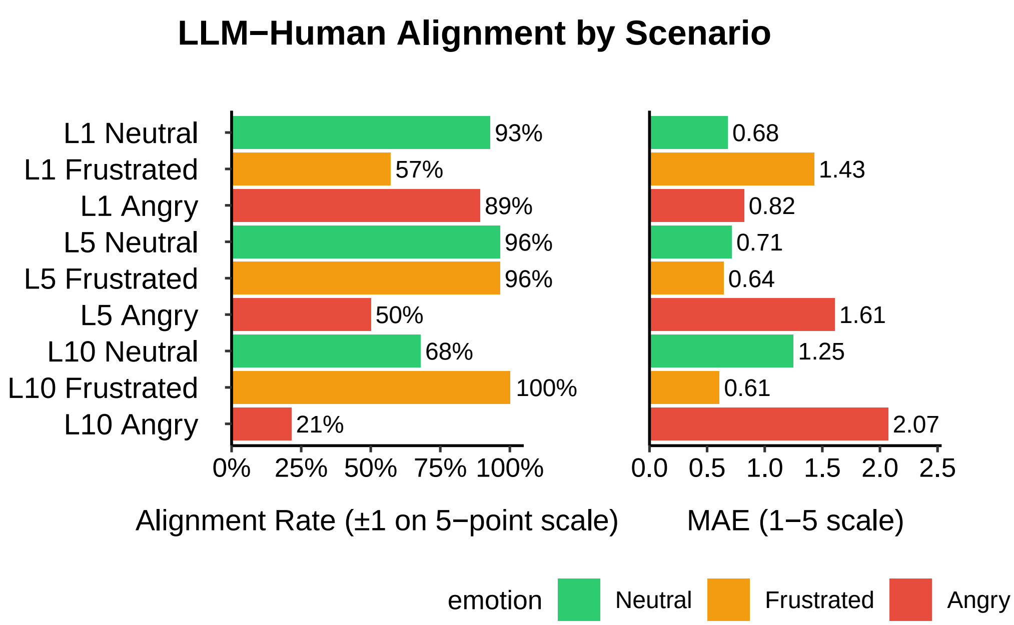

Figure 3: LLM–Human Alignment by Scenario

0.3.3 LLM & Human Alignment Broken Down to Emotion & Topic Intensity

Table 5: LLM-Human Alignment Statistics by Emotion

level	alignment_rate	Spearman_rho	MAE
L1	0.7976190	-0.0054286	0.9761905
L5	0.8095238	0.1476211	0.9880952
L10	0.6309524	0.1252275	1.3095238

Table 6: LLM-Human Alignment Statistics by Topic Intensity

emotion	alignment_rate	Spearman_rho	MAE
angry	0.5357143	0.0240584	1.5000000
frustrated	0.8452381	0.0338909	0.8928571
neutral	0.8571429	-0.0795702	0.8809524

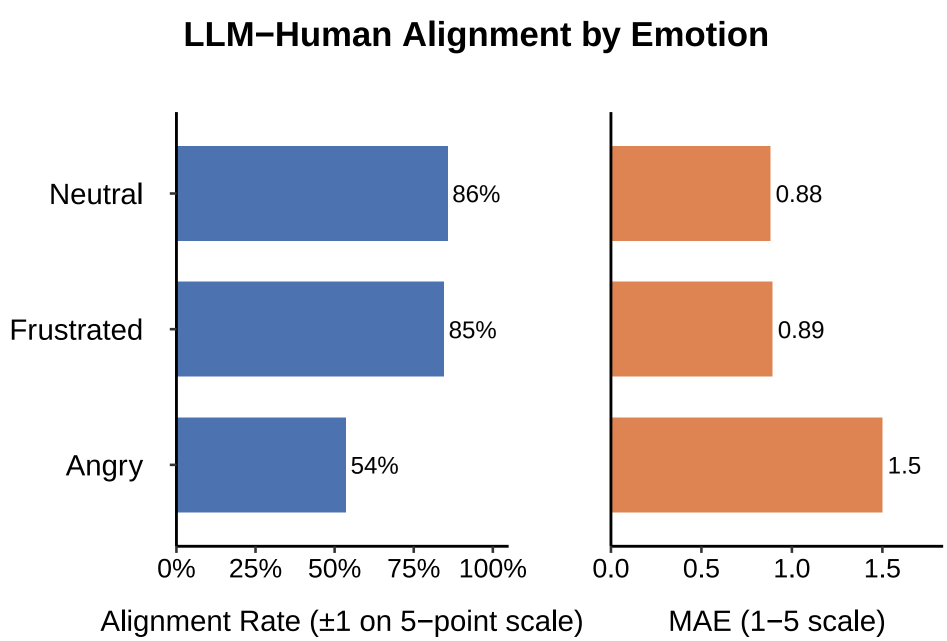

Figure 4: LLM–Human Alignment by Emotion

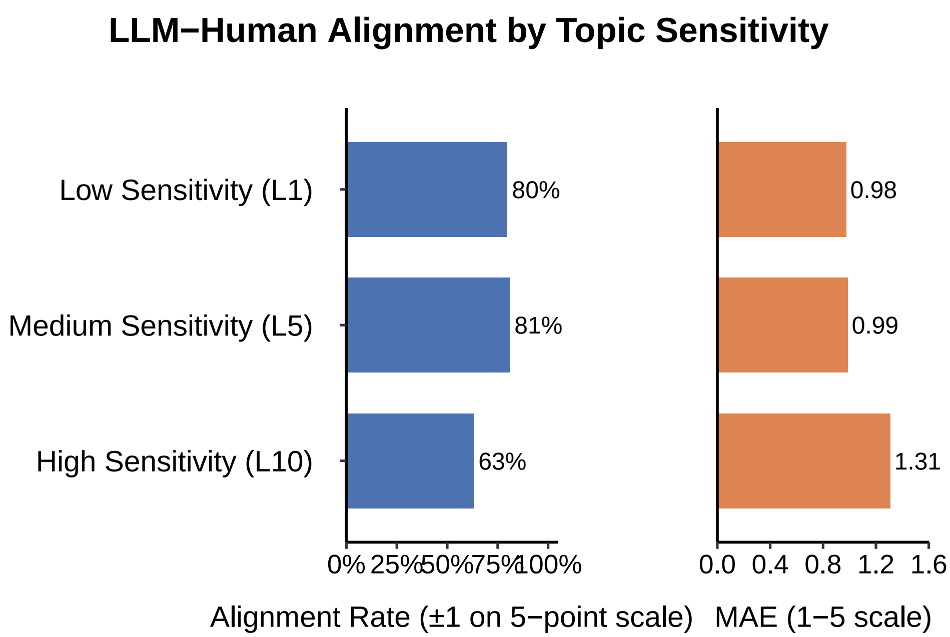

Figure 5: LLM–Human Alignment by Topic Sensitivity